Topic 29 - Sub-Workflows, Debugging & Production Readiness

Step 1: Build Child Sub-Workflow 1 - Tax Calculation Engine

1. Create Sub-Workflow:

In your n8n dashboard, click **+ Add workflow**.

Rename in top-left to: `Sub-Workflow 1 - Tax Engine`.

2. Add Execute Workflow Trigger:

Click **+ Add first step...**, search for **`Execute Workflow Trigger`**, and place it. Rename to: `Execute_Workflow_Trigger`.

3. Add State Tax Calculation Engine:

From `Execute_Workflow_Trigger`, click **`+`**, search for **`Code`**, and add the node. Rename to: `Code_CalculateStateTax`.

Language: Select **`JavaScript`**.

Mode: Select **`Run Once for Each Item`**.

JavaScript Code: Paste:

```
const order = $input.item.json;
const stateRates = { 'CA': 0.0825, 'NY': 0.08875, 'TX': 0.0625, 'FL': 0.0600 };



const rate = stateRates[order.state] || 0.0500; // 5% default



const subtotal = Number(order.subtotal || 0);
const taxAmount = Number((subtotal * rate).toFixed(2));
const grandTotal = Number((subtotal + taxAmount).toFixed(2));

return {
 json: {
 orderId: order.orderId,
 subtotal: subtotal,
 state: order.state,
 appliedTaxRate: rate,
 taxAmount: taxAmount,
 grandTotal: grandTotal,
 taxCalculatedAt: new Date().toISOString()
 }
};
```

4. **Save & Copy Workflow ID:**

   Click **Save** in the top-right toolbar.

   Note the URL in your browser bar (e.g. `http://.../workflow/<subworkflow-id-1>`). Copy this ID.

Step 2: Build Child Sub-Workflow 2 - Payment Gateway Processor

1. **Create Sub-Workflow:**

   Click **+ Add workflow**. Rename to: `Sub-Workflow 2 - Payment Gateway`.

2. **Add Trigger & Payment Execution:**

   Add **Execute Workflow Trigger** node named `Execute_Workflow_Trigger`. From trigger, add **HTTP Request** node named `HTTP_ProcessPaymentCharge`. **Method:** Select **`POST`** (Mode: **Fixed**).

   URL: Toggleto **Expression (`fx`)**, enter: `https://dummyjson.com/c/<your-id>` (or mock payment endpoint).

   Send Body: Switch toggle to **`ON`**.

   Body Content Type: Select **`JSON`**.

   Specify Body: Select **`Using JSON`**.

   JSON: Switch to **Expression (`fx`)** and enter:

   ```
   {{ { "orderId": $json.orderId, "amount": $json.grandTotal } }}
   ```

3. **Add Response Formatter:**

   Connect to a **Code** node named `Code_FormatPaymentReceipt`:

   ```
   const resp = $input.item.json;

   return {
    json: {
    paymentStatus: "CONFIRMED",
    transactionRef: "AUTH-" + Math.random().toString(36).substring(2,
   9).toUpperCase(),  amountCharged:
   $node['Execute_Workflow_Trigger'].json.grandTotal,
    processedAt: new Date().toISOString()
    }
   };
   ```

4. **Save & Copy Workflow ID:**

   Click **Save** and note `<subworkflow-id-2>`.

Step 3: Build the Master Order Orchestrator Workflow

1. **Create Master Workflow & Trigger:**

   Click **+ Add workflow**, rename to: `Topic 14 - Sub-Workflows, Debugging & Production Readiness`.

   Add a **Manual Trigger** node named `Manual_Trigger`.

2. **Add Master Order Stager Node:**

   From `Manual_Trigger`, click `+`, search for **`Code`**, and add the node.

   Rename to: `Code_StageMasterOrder`.

   Language: Select **`JavaScript`**.

   Mode: Select **`Run Once for All Items`**.

   JavaScript Code: Paste:

```
return [
 {
 json: {
 orderId: "ORD-MODULAR-901",
 customerName: "Acme Corporation",
 state: "CA",
 subtotal: 1000.00
 }
 }
];
```

3. **Test Node Checkpoint:**

   Click **Test step**. Verify staged order appears in Output.

Step 4: Invoke Tax Sub-Workflow (**`Execute_TaxSubWorkflow`**)

1. **Add Node to Canvas:**

   From `Code_StageMasterOrder`, click `+`, search for **`Execute Workflow`**, and select the **Execute Workflow** node.
   Rename to: `Execute_TaxSubWorkflow`.

2. **Configure Parameters (Parameters Tab):**

   Source: Select **`Database`** (or `From list`).

   Workflow: Select `Sub-Workflow 1 - Tax Engine` (or enter `<subworkflow-id-1>`). **Mode:** Select **`Once for each item`**.

3. **Test Node Checkpoint:**

   Click **Test step**. Notice that the TaxSub-Workflow runs remotely and returns `appliedTaxRate: 0.0825`, `taxAmount: 82.50`, and `grandTotal: 1082.50`!

Step 5: Merge Tax Outputs (`Code_MergeTaxTotals`)

1. **Add Node to Canvas:**
   From `Execute_TaxSubWorkflow`, click `+`, search for `Code`, and add the node. Rename to: `Code_MergeTaxTotals`.
2. **Configure Parameters (Parameters Tab):**
   Language: Select `JavaScript`.
   Mode: Select `Run Once for Each Item`.
   JavaScript Code: Paste:

```
const order = $node['Code_StageMasterOrder'].json;
const taxResult = $input.item.json;

return {
 json: {
 ...order,
 taxRate: taxResult.appliedTaxRate,
 taxAmount: taxResult.taxAmount,
 grandTotal: taxResult.grandTotal
 }
};
```

3. **Test Node Checkpoint:**
   Click **Test step**. Verify merged order contains `grandTotal: 1082.5`.

Step 6: Invoke Payment Sub-Workflow (`Execute_PaymentSubWorkflow`)

1. **Add Node to Canvas:**
   From `Code_MergeTaxTotals`, click `+`, search for `Execute Workflow`, and add the node. Rename to: `Execute_PaymentSubWorkflow`.
2. **Configure Parameters (Parameters Tab):**
   Source: Select `Database`.
   Workflow: Select `Sub-Workflow 2 - Payment Gateway` (or enter `<subworkflow-id-2>`). **Mode:** Select `Once for each item`.
3. **Test Node Checkpoint:**
   Click **Test step**. Verify the payment sub-workflow executes and returns `paymentStatus: "CONFIRMED"`.

Step 7: Build Final Orchestrator Summary (`Code_BuildOrchestratorSummary`)

1. **Add Node to Canvas:**

From `Execute_PaymentSubWorkflow`, click `+`, search for **`Code`**, and add the node. Rename to: `Code_BuildOrchestratorSummary`.

2. **Configure Parameters (Parameters Tab):**

   Language: Select **`JavaScript`**.

   Mode: Select **`Run Once for Each Item`**.

   JavaScript Code: Paste:

```
const payment = $input.item.json;
const order = $node['Code_MergeTaxTotals'].json;

return {
 json: {
 orchestrationResult: "ORDER_FULFILLED_VIA_MICROSERVICES",
 orderId: order.orderId,
 customer: order.customerName,
 grandTotalPaid: order.grandTotal,
 authCode: payment.transactionRef,
 executedSubWorkflows: [
 "Sub-Workflow 1: Tax Engine",
 "Sub-Workflow 2: Payment Gateway"
 ],
 completedAt: new Date().toISOString()
 }
};
```

3. **Test Node Checkpoint:**

   Click **Test step**. Verify consolidated master order execution receipt.